

Intro to SQL

 <https://github.com/learn-co-curriculum/python-p3-intro-to-sql>  <https://github.com/learn-co-curriculum/python-p3-intro-to-sql/issues/new>

Learning Goals

- Use SQL to store data and retrieve it later on.
- Use SQLite to build relational databases on your computer.
- Perform CRUD operations on relational databases using SQL.

Key Vocab

- **SQL (Structured Query Language)**: a programming language that is used to manage relational databases and perform operations on the data that they contain.
- **Relational Database**: a collection of data that is organized in well-defined relationships. The most common type of database.
- **Query**: a statement used to return data from a database.
- **Table**: a collection of related data in a database. Composed of rows and columns. Similar to a class in Python.
- **Row**: a single record in a database table. Each column represents an attribute of the record. Similar to an object in Python.
- **Column**: a single field in a database table. Each row contains values in each column. Similar to a Python object's attributes.
- **Schema**: a blueprint of the construction of the tables in a database and how they relate to one another.

Introduction

Just as humans have brains, and computers have hard drives that store information, most programs have a storage system that needs to be communicated with. How do we store and find stored data? In databases, and using *Structured Query Language*, or SQL. SQL can be used with a ton of different databases like MySQL, Postgres, etc. It powers the logic behind server-side frameworks. SQL has many variations, but for our

lessons, we will be using [SQLite](https://www.sqlite.org/index.html)  [_\(https://www.sqlite.org/index.html\)_](https://www.sqlite.org/index.html). While there are some minor differences we'll note later on, the fundamentals stay the same.

What is SQL?

SQL (Structured Query Language), pronounced like "sequel" or "S-Q-L" depending who you ask, is a language for managing data in a database. Unlike some other programming languages, it's only used for one thing: talking to databases. Thus, you might hear it referred to as a "special purpose", or "domain specific" programming language. This means that you won't be using SQL alone to write the next big web app, but you might use it to interact with the database that powers it.

Even though SQL has just one purpose, it is used by many different database systems such as MySQL, PostgreSQL, or the system we'll be using in this course: SQLite. There are a handful of things other SQL systems like Postgres, or MySQL can do that are not supported by SQLite. Every database system has its own strengths and weaknesses, and as you learn more about them you should evaluate them thoughtfully when deciding which to use for what purpose. For us, SQLite provides a low barrier to entry, and is simple to get up and running.

Web Developers <3 Databases

Why do we need to learn SQL? As a full-stack web developer, you'll frequently be working with databases to manage the data associated with your applications. Think about the web apps you're familiar with. Facebook saves user data and stores — or persists — your associations to your friends. Amazon has an enormous database of items for sale. The New York Times has a storage system for all of their articles. Most of the sites you interact with every day, and consequently most of the sites you'll work on or build, need databases to persist data.

We've already seen how we can build Python programs that model real-world objects and environments. We've seen one form of "persistence" for our objects by storing objects in memory when our applications are running, like by using the class attributes to store all instances of the class that are created when our application runs.

With database management skills, we'll learn to store representations of the Python objects our programs create and retrieve them at the appropriate time. We'll also learn how to connect our Python (and even later, Flask) applications to our databases.

For example, a basic web application might have many users. So far, we've learned how to build a Python `User` class that produces user objects. But, we don't yet know how to store those users and their details. If a user signs up for our app and we proceed to lose all their information immediately, how will we know if an existing user is signing back into our app? We need a way to take our Python objects, store them in a database and retrieve them at the appropriate time.

In This Section

This section will cover how to use and navigate databases based on SQL. By the end of this section, you'll be able to:

- Create SQLite3 databases
- Create, update, select, and delete data from database tables
- Relate data within a given database
- Write SQL code in both your command line and your text editor and execute the code against a database
- Write Python programs that talk to and save data to your databases

IMPORTANT NOTE: This section introduces a lot of new SQL syntax — you're being introduced to an entirely new programming language! However, a bit later in this phase, we'll be introducing a tool, SQLAlchemy, which will allow us to use Python classes to interact with a SQL database instead of writing SQL ourselves.

It's important to still have an understanding of what SQLAlchemy does for us under the hood, and you'll need to understand some SQL syntax to know what SQLAlchemy is actually doing, and how to debug some common issues and improve your code's performance.

That being said: don't worry too much about memorizing all the *syntax* in this section. Work through the lessons, but move on if you get stuck. It's more important that you learn the *concepts* from this section than that you gain mastery over writing SQL.

Resources

- [SQL Tutorial - W3Schools](https://www.w3schools.com/sql/) ➞ [\(https://www.w3schools.com/sql/\)](https://www.w3schools.com/sql/)
- [Documentation - SQLite](https://www.sqlite.org/docs.html) ➞ [\(https://www.sqlite.org/docs.html\)](https://www.sqlite.org/docs.html)
- [SQLite - VisualStudio Marketplace](https://marketplace.visualstudio.com/items?itemName=alexcvzz.vscode-sqlite) ➞ [\(https://marketplace.visualstudio.com/items?itemName=alexcvzz.vscode-sqlite\)](https://marketplace.visualstudio.com/items?itemName=alexcvzz.vscode-sqlite)

- [SQLite Keywords - SQLite](https://www.sqlite.org/lang_keywords.html)  [\(https://www.sqlite.org/lang_keywords.html\)](https://www.sqlite.org/lang_keywords.html)
- [ZetCode sqlite3 Tutorial](http://zetcode.com/db/sqlite/)  [\(http://zetcode.com/db/sqlite/\)](http://zetcode.com/db/sqlite/)